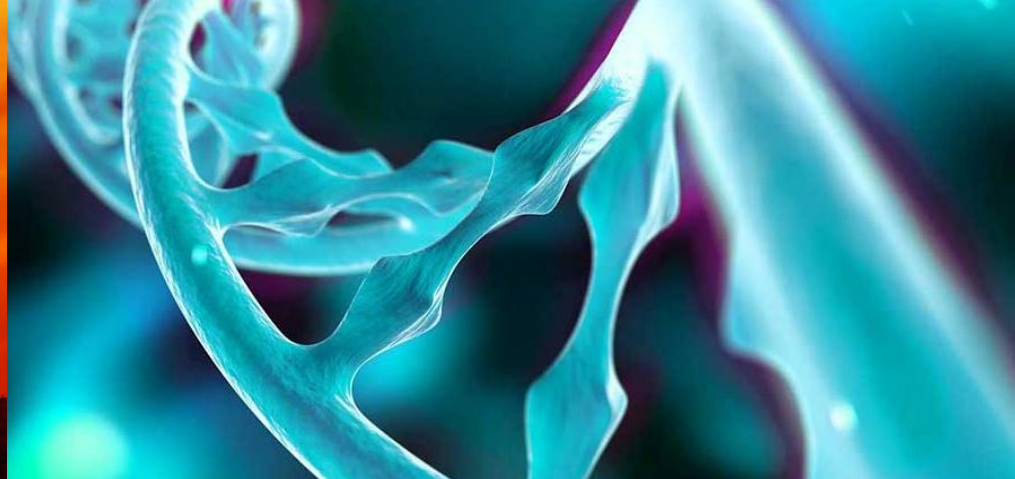
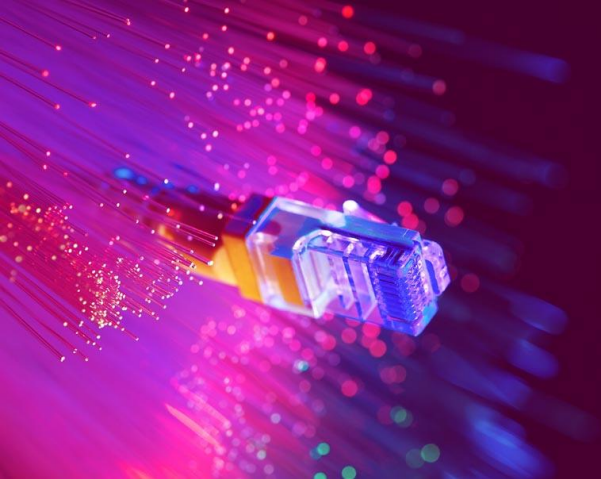




Bioinformatics data analysis using Chipster



CSC – Suomalainen tutkimuksen, koulutuksen, kulttuurin ja julkishallinnon ICT-osaamiskeskus

You will learn



- What Chipster is
- How the user interface works
- How to import data
- How to store and share your analysis
- How to speed up your analysis
 - run the same analysis step for all the samples simultaneously
- How to get help
- Where to find training materials

Chipster: User-friendly analysis software for high-throughput data



- Over 500 analysis tools, inc tools for single-cell RNA-seq and Visium data
 - Command line tools, R/Bioconductor packages
 - Free, open source software
- Users can share analysis sessions, analysis metadata is tracked
- Training resources
 - Course material (slides, videos, exercises, data sets) available
 - Training accounts available, email chipster@csc.fi
- Technical
 - Angular based web app and cloud native scalable backend
 - Ansible playbooks and Helm templates for setting up a K3s container orchestration system and Chipster containers on a virtual machine or on a physical server
- <https://chipster.csc.fi/>



Chipster

Open source platform for data analysis



- Home
- Getting access
- Manual
- Tutorial videos
- Course material
- Cite
- Contact

Welcome to Chipster

Chipster is a user-friendly analysis software for high-throughput data such as Visium, single-cell and bulk RNA-seq. Chipster provides a web interface to over 500 analysis tools, and the actual analysis jobs run on the server side making use of CSC's computing environment (watch [introduction video](#)).

If you would like to use Chipster hosted by CSC, you need a [user account](#). Please note that Chipster is also available for [local server installation](#) free of charge.



Launch Chipster

Training:

- 3.-4.3.2026 [Single-cell RNA-seq data analysis](#)
- 14.-15.4.2026 [ASV-based microbial community / environmental DNA analysis](#)
- [MOOC Single-cell RNA-seq data analysis using Chipster](#)
- [MOOC Spatially resolved transcriptomics data analysis using Chipster](#)

News and resources:

- [video tour of Chipster Web app](#)
- ASV-based microbial community analysis using DADA2: [Tutorial videos](#)
- [Instructions for moving data from Puhti to Chipster](#)
- [Video on how to convert tables to Chipster format and create phenodata file](#)

Chipster user interface (chipster.2.rahtiapp.fi)



Chipster **Analyze** Sessions Manual Getting Access Contact News

ekorpela@csc.fi

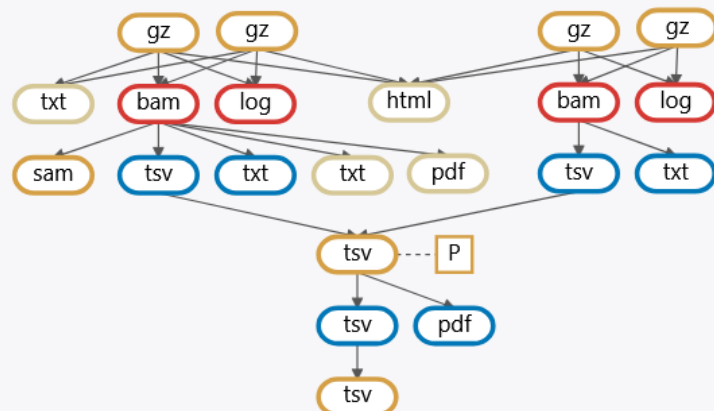
Files

Workflow

List

Find file

Add file



Tools

Toolset

NGS

Category

- Quality control
- Preprocessing
- Utilities
- Matching sets of genomic regions
- Alignment
- Variants
- Small RNA-seq
- RNA-seq
- Single-cell RNA-seq (Seurat v5)
- Spatially resolved transcriptomics (Seurat v5)
- ChIP- and DNase-seq
- Microbial amplicon data preprocessing for ASV
- Microbial amplicon data preprocessing for OTU
- Microbial amplicon data analyses
- CNA-seq

Tool

Read quality with MultiQC for many FASTQ files

- Read quality with FastQC
- Read quality statistics with FASTX
- RNA-seq quality metrics with RSeQC
- RNA-seq strandedness inference with RSeQC
- Collect multiple metrics for BAM
- PCA and heatmap of samples with DESeq2
- Check FASTQ file for errors

Parameters

Run

The tool runs FastQC on multiple FASTQ files, and then combines the reports using MultiQC. Input can be FASTQ files or tar files containing FASTQ files. Files can be gzipped. Please make sure you don't have duplicate FASTQ file names. Run the tool once for all samples, not separately for each file. This tool is based on the FastQC and MultiQC packages. [More info...](#)

Session Info

04_RNAseq_lung_lymphnode_comparison_2samples ...

Created: 3/2/2026 11:06:39 AM

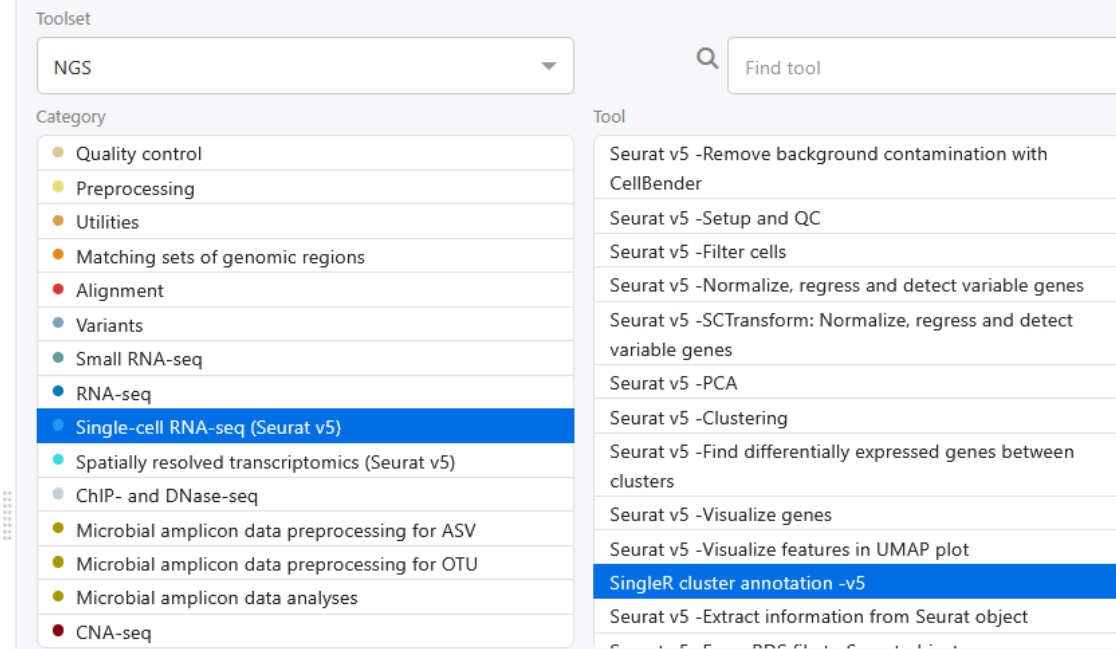
Size: 116 MB

In this tutorial we compare gene expression in human lung and lymph node samples. To make things faster, the data is given in two separate sessions:

-In this session we have two paired-end samples in FASTQ format. In order to make the analysis faster during the course, only a small subset of the reads (200 000) is used. You will preprocess and align the reads, make a count table and perform differential expression analysis.

-In the second session (RNAseq_lung_lymphnode_comparison_10samples) we have a ready-made count table, generated from 10 full size samples. We will perform experiment level quality control, differential expression analysis, filtering and annotation.

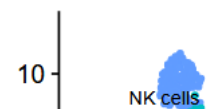
Tools



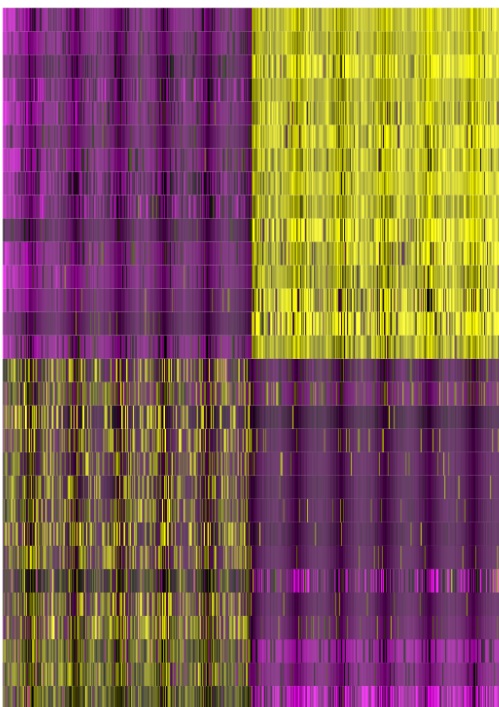
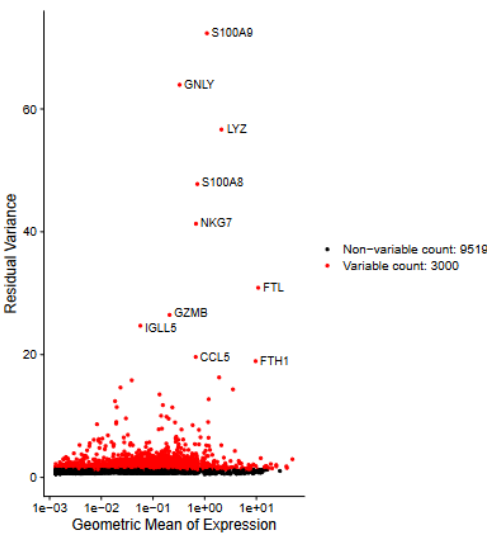
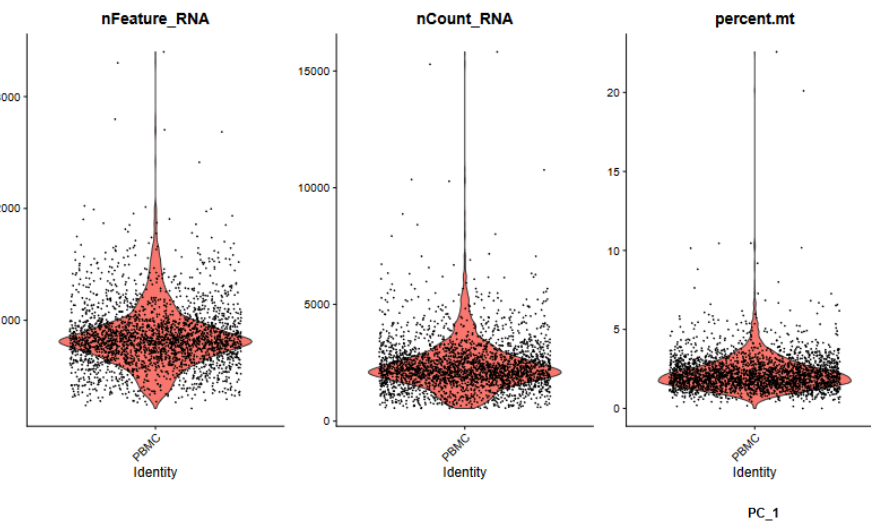
singleR_annotation_plots.pdf ...

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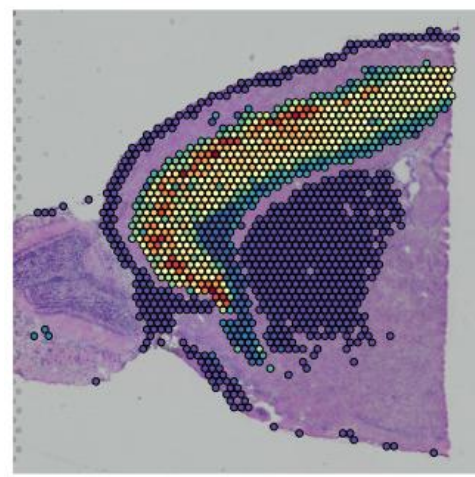
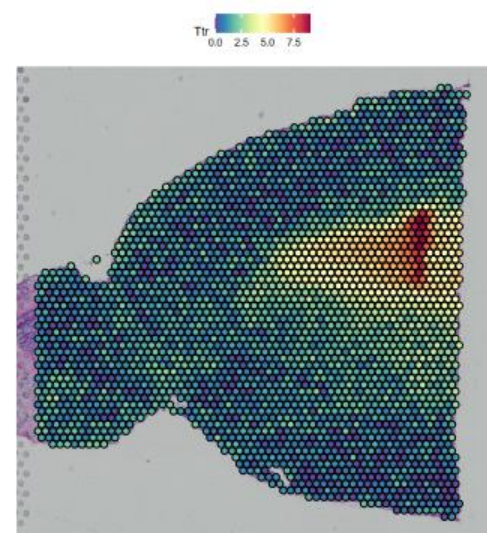
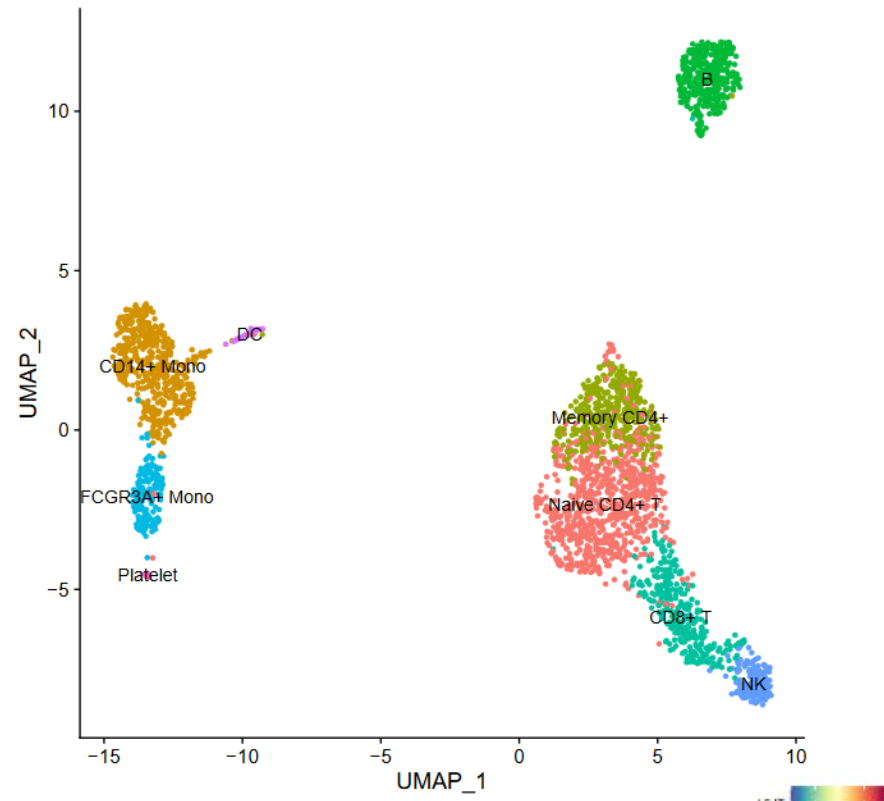
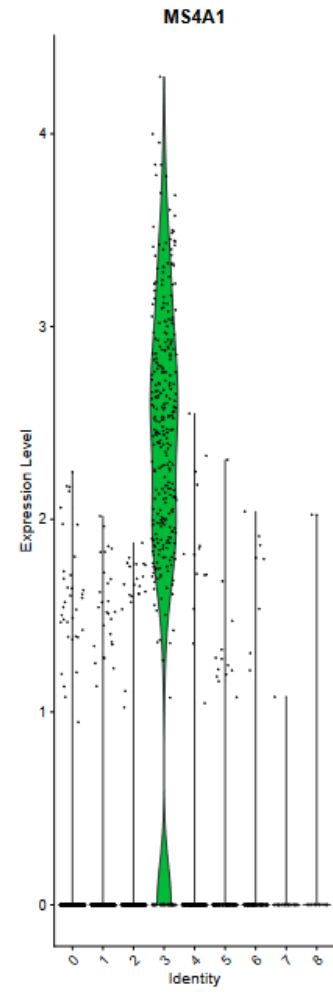
Main annotations based on clusters



Chipster visualizations



GST3
TYROBP
S100A9
PTL
LST1
FCN1
AIF1
LYZ
FTH1
S100A8
TYMP
FCER1G
CFD
LGALS2
LGALS1
CCL5
TRAF3IP3
AQP3
GIMAP5
STK17A
CD247
CTSW
CD27
ACAP1
B2M
CD2
IL7R
LTB
IL32
MALAT1



Options for importing data to Chipster



- “Add file” button
 - Upload files
 - Upload folder
 - Download from URL
- Sessions tab
 - Import session file
- Tools
 - Import from Illumina BaseSpace
 - Utilities / Retrieve data from Illumina BaseSpace
 - Access token needed
 - Import from SRA database
 - Utilities / Retrieve FASTQ or BAM files from SRA
 - Import from Ensembl database
 - Utilities / Retrieve data for a given organism in Ensembl
 - Import from URL
 - Utilities / Download file from URL directly to server

Analysis sessions

- Your analysis is saved automatically in the cloud
 - Session includes all the files, their relationships and metadata (what tool and parameters were used to produce each file).
 - Session is a single .zip file.
 - Note that cloud sessions are not stored forever! Remember to download the session when ready.
- You can share sessions with other Chipster users
 - You can give either read-only or read-write access
- If your analysis job takes a long time, you don't need to keep Chipster open:
 - Wait that the data transfer to the server has completed
 - Close Chipster
 - Open Chipster later and the results will be there

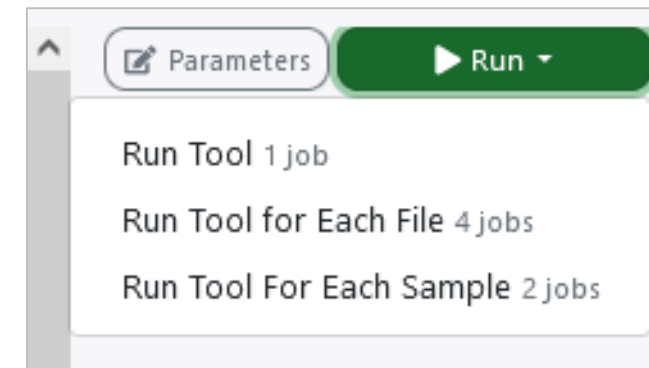
Running many analysis jobs at the same time



- You can have many analysis jobs running at the same time
 - No need to wait that one finishes before starting a new one

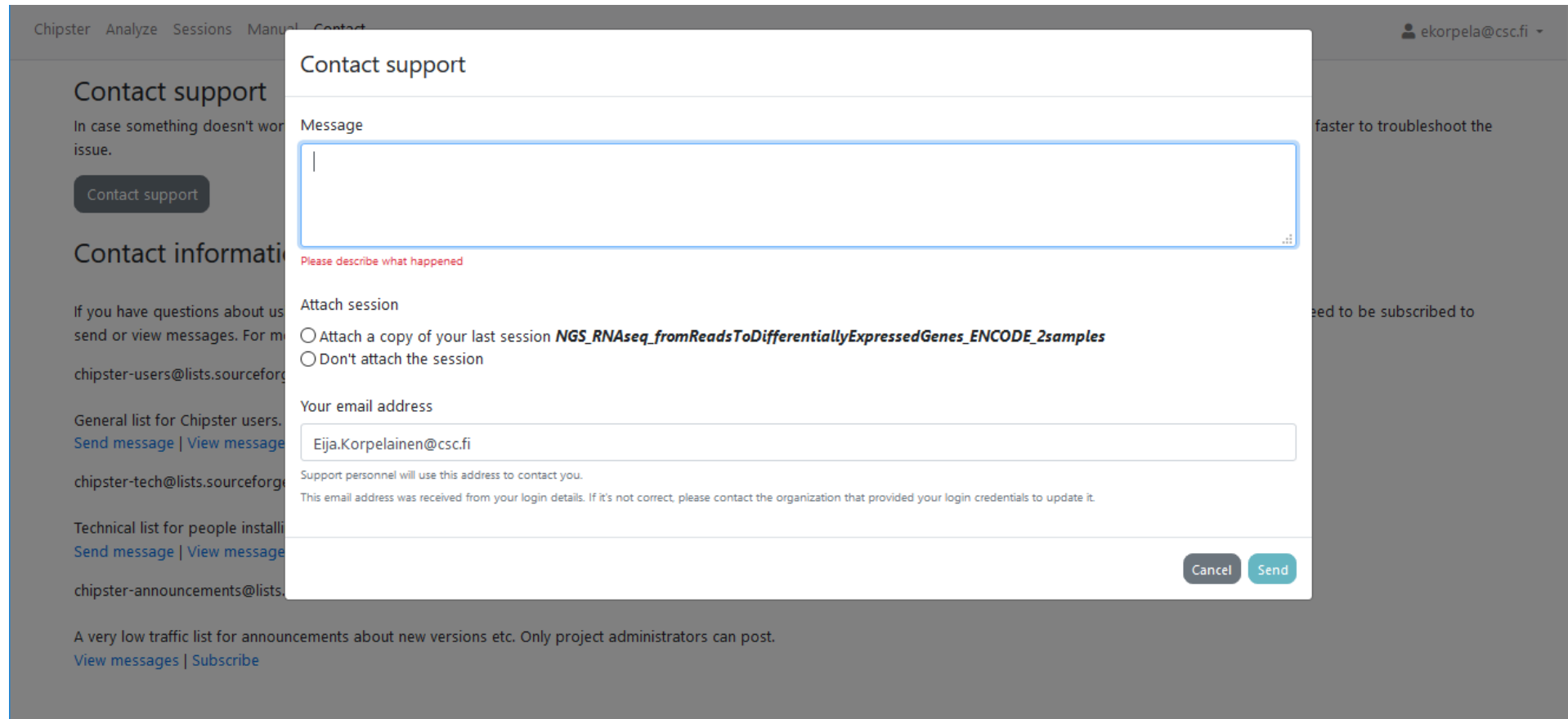
Run button gives several options:

- Run tool
 - Runs the selected analysis tool once
- Run tool for each file
 - Runs the selected analysis tool for each of the input files individually
- Run tool for each sample
 - If you have grouped paired end FASTQ files to samples using the Define samples –option, you can run the selected analysis tool for the input files in a sample specific manner.



Problems? Send us a support request

-request includes the error message and link to analysis session (optional)



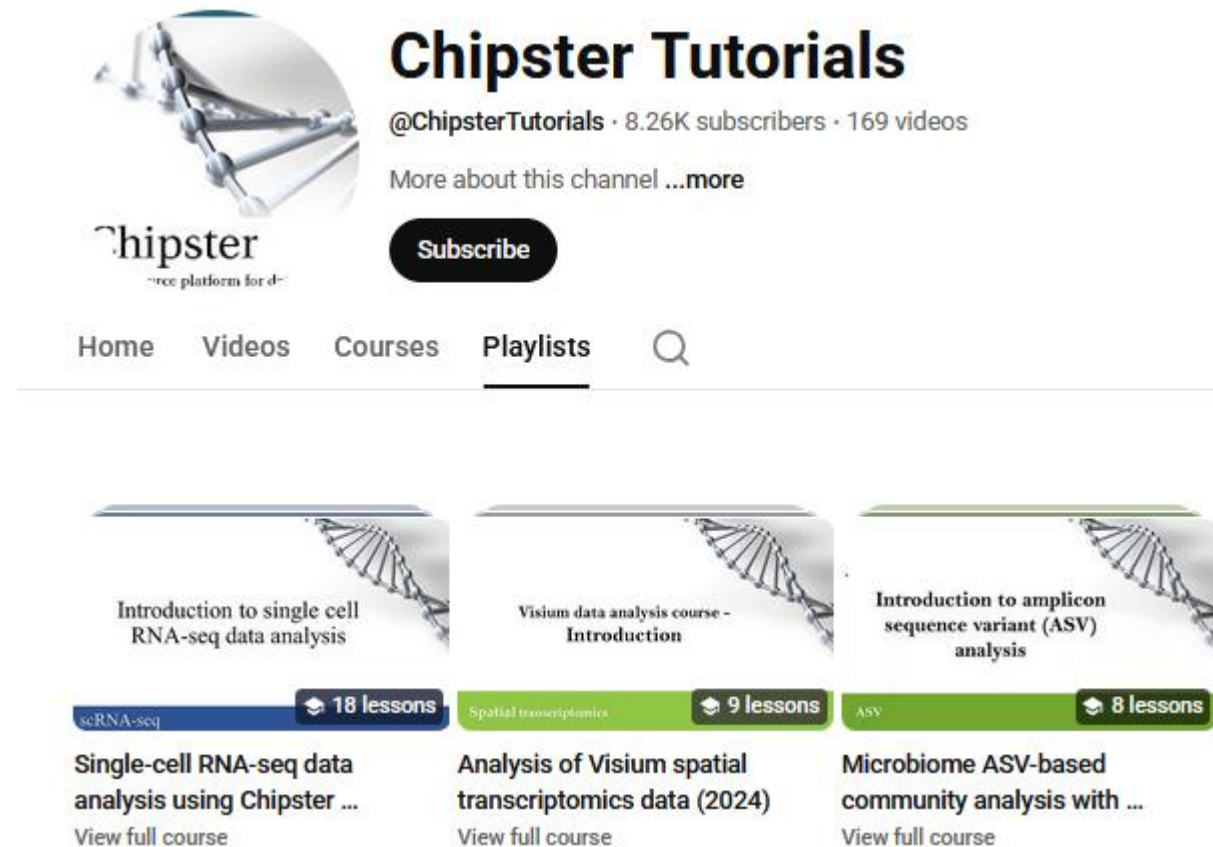
The screenshot shows the Chipster web interface with a 'Contact support' modal form open. The background page has a navigation bar with 'Chipster', 'Analyze', 'Sessions', 'Manual', and 'Contact'. The user is logged in as 'ekorpela@csc.fi'. The modal form is titled 'Contact support' and contains the following fields and options:

- Message:** A large text area for describing the issue. Below it is a red prompt: "Please describe what happened".
- Attach session:** Two radio button options:
 - ☐ Attach a copy of your last session *NGS_RNAseq_fromReadsToDifferentiallyExpressedGenes_ENCODE_2samples*
 - ☐ Don't attach the session
- Your email address:** A text field containing 'Eija.Korpelainen@csc.fi'. Below the field is a note: "Support personnel will use this address to contact you. This email address was received from your login details. If it's not correct, please contact the organization that provided your login credentials to update it."
- Buttons:** 'Cancel' and 'Send' buttons at the bottom right of the modal.

The background page also shows 'Contact support' and 'Contact information' sections with links to various mailing lists and a 'Contact support' button.

More info

- chipster@csc.fi
- <http://chipster.csc.fi>
- <https://chipster.2.rahtiapp.fi/manual/courses.html>
- Chipster tutorials in YouTube



The screenshot shows the YouTube channel page for "Chipster Tutorials". The channel name is "Chipster Tutorials" with the handle "@ChipsterTutorials", 8.26K subscribers, and 169 videos. Below the channel name is a "Subscribe" button and a link to "More about this channel ...more". The navigation bar includes "Home", "Videos", "Courses", "Playlists", and a search icon. The "Courses" tab is selected. Three course cards are displayed:

- Introduction to single cell RNA-seq data analysis** (18 lessons)
Single-cell RNA-seq data analysis using Chipster ...
View full course
- Visium data analysis course - Introduction** (9 lessons)
Analysis of Visium spatial transcriptomics data (2024)
View full course
- Introduction to amplicon sequence variant (ASV) analysis** (8 lessons)
Microbiome ASV-based community analysis with ...
View full course